# How Much Targeted Training Does It Take to Top a Multi-Task Leaderboard? Minimum Investment Under Different Aggregation Rules

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## Abstract

A leaderboard turns per-task scores into one rank, so a developer can try to lift a chosen model by training on data related to individual sub-datasets. How much such targeted training is needed to put the model first, and how does that depend on the aggregation rule? We define the investment as the minimum raise of the target's task scores, within a per-task cap, that makes it strictly first, and measure it in points, tasks touched and largest single-task jump. We solve it exactly for the mean, median, minimum, Borda and Condorcet-winner rules, prove closed forms for the first three, and apply it to every model of the Open LLM Leaderboard (4495 targets, 6 tasks) and of HELM MMLU (78 targets, 57 subjects). With caps taken from observed fine-tuning gains, first place is reachable for only 12% of OLLB targets under the mean (largest feasible rank 901) and 19% of HELM targets (largest feasible rank 16); with a 100-point cap every target can reach it (cap sweep: appendix Table 4 and Figures 4–5). Among feasible targets the mean is the costliest rule and the median and minimum the cheapest: at OLLB ranks 11–50 the other rules need 0.18–0.50 of the mean's total points. Observed costs lie inside the 95% intervals from evaluation noise in all 66 feasible target–rule cases checked (appendix Figure 6), but whether a target near the frontier is feasible at all is noise-sensitive. Conclusions depend on the cap, which on HELM is transferred from OLLB, and hold for fixed rivals and score tables only.

## 1 Introduction

Multi-task benchmarks rank models by aggregating per-task scores into one number, and the resulting rank influences which models are adopted, published and funded. A developer therefore has an incentive to improve a chosen target model specifically on the benchmark's sub-datasets, for instance by training on data related to them. How much such targeted improvement is needed to reach first place, and how that amount depends on the aggregation rule, is the question we study on public leaderboard data.

Our approach is deliberately simple and exact. We take the score table of a public leaderboard as fixed, let the target raise its own scores task by task up to a cap, and compute the *minimum* raise that makes it strictly first under a given aggregation rule. The cap is the central modelling assumption: it stands for how far training on related data can move one score. We therefore report results under a data-grounded cap and an oracle cap, and report infeasibility (no allowed raise makes the target first) as an outcome rather than a missing value.

**Contributions.** (i) A formulation of the minimum targeted investment with three cost measures ($L_1$, $K$, $L_\infty$) and an explicit family of caps (Section 3). (ii) Exact solvers for score rules (mean,

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median, minimum, power and trimmed means) and rank rules (Borda, Condorcet winner, plurality, Copeland), tested against brute force, with closed forms and proofs for the mean, minimum and median (Section 3.4, Appendix A). (iii) A study of every model on two public leaderboards, with bootstrap and evaluation-noise intervals (Sections 4–5).

**Findings.** Under caps grounded in observed fine-tuning gains, first place is out of reach for most models, under every rule: the feasible fraction is 12% (mean) on OLLB and 19% on HELM, and the rule changes how far down the board it is reachable. Where it is reachable, the mean rule costs the most points, and order-statistic and rank rules cost between 18% and 50% of that at OLLB ranks 11–50. Two of three hypotheses fixed beforehand are only partly borne out (Section 5). All conclusions are conditional on the cap and on the other models being fixed; Section 6 lists where they fail.

## 2 Related work

**Benchmarks and leaderboards.** MMLU [Hendrycks et al., 2020] aggregates 57 subjects, and HELM [Liang et al., 2022] releases per-scenario scores and raw model outputs for many models; BIG-Bench Hard [Suzgun et al., 2022] selects hard tasks from BIG-Bench. These provide the multi-task score matrices that we analyse.

**Fragility of rankings.** Dehghani et al. [2021] show that the relative order of methods can change with the choice of benchmark tasks (the "benchmark lottery"). Singh et al. [2025] document how private testing and unequal data access distort the Chatbot Arena leaderboard. Both concern what is measured and disclosed; we instead hold the leaderboard fixed and ask what a single developer must change for one model to win.

**Aggregation and social choice.** Lanctot et al. [2023] treat each task as a voter and use voting theory to build cross-task evaluation; the Handbook of Computational Social Choice [Brandt et al., 2016] covers the rules (Borda, Condorcet-consistent rules, plurality) we reuse. Our rules are the same families, but our question is the cost of changing the winner, not the choice of rule.

**Training on the test task and contamination.** Zhou et al. [2023] show that benchmark leakage into training can raise scores sharply, and Yang et al. [2023] show that rephrased test samples evade common decontamination. Chen et al. [2026] analyse, in a game-theoretic model, the incentives that benchmarks create for strategic post-training. Our analysis is empirical and combinatorial: given a fixed score table, we compute the exact minimum targeted improvement per rule, under explicit assumptions about how much any one task can be improved.

**Positioning.** Closest in question is Chen et al. [2026], who model developers choosing how much to spend on benchmark-specific improvement of a single benchmark score, prove non-existence of equilibria for current protocols and a unique equilibrium for tune-before-test. That analysis concerns one score per model; the choice of aggregation rule over tasks does not enter it. We take the opposite stance: no game and no equilibrium, a fixed table, and the exact cost for one developer under each of several aggregation rules. Colombo et al. [2022] propose a social-choice-based aggregation for NLP benchmarks and show it changes conclusions relative to the mean; we do not propose a rule but measure how costly it is to move a model to first place under each. The computational-social-choice literature studies the analogous *bribery* problem, the cost of changing votes so that a chosen candidate wins under a voting rule [Faliszewski et al., 2009, Brandt et al., 2016]. Our setting differs in that the manipulator changes its own scores on tasks (the "voters") and pays in score points under per-task caps, which gives closed forms for score rules and integer programs for rank rules.

## 3 Problem setup

### 3.1 Leaderboard, target and investment

A leaderboard is a score matrix $X \in [0, 100]^{M \times n}$ of $M$ models on $n$ tasks. Fix a target model $i$ with score vector $x = X_{i\cdot}$; all other rows stay fixed. In targeted training the developer can raise the

target's score on task $t$ to any $y_t \in [x_t, c_t]$, where $c_t$ is a *cap* (Section 3.2). An aggregation rule $f$ maps the score vectors to a ranking; the target is *first* if it is strictly above every other model under $f$ (by a margin $\varepsilon = 10^{-3}$ points for score rules). We report the investment in three ways:

$$L_1 = \sum_t (y_t - x_t), \qquad K = \#\{t : y_t > x_t\}, \qquad L_\infty = \max_t (y_t - x_t),$$

the total points, the number of tasks touched, and the largest single-task jump. The minimum of each is taken separately; $L_\infty$ is also the cost of the uniform "general improvement" that raises every task by the same amount. For comparison across rules and datasets we also report $K/n$ and $L_1/n$.

### 3.2 Caps: how much can one task be improved?

Without a bound on $y_t$ the question is trivial (put every task at 100). The cap $c_t$ is therefore the central modelling assumption, and we never collapse it to a single choice. Two caps appear in the main text: `ceil`, $c_t = 100$, an oracle upper bound on what can be done and so a lower bound on any required investment; and `emp99`, $c_t = \min(100, x_t + g_t^{99})$ where $g_t^{99}$ is the 99th percentile of the observed gains (fine-tune minus its declared base model) on task $t$ in the Open LLM Leaderboard. `emp99` is the only data-grounded cap. It inherits a selection bias, since only published fine-tunes appear, and it measures gains attained by the community, not the gains a determined developer could attain. Further caps (best existing score $+0$ or $+2$, and at most $+5/+10/+20$ points per task) are reported in the appendix. When no $y \le c$ makes the target first, the case is *infeasible*; infeasibility is reported as a rate and never coded as a large investment, and all investment statistics are conditional on feasibility.

### 3.3 Aggregation rules

Score rules apply a function $f$ to the score vector of each model: the *mean* (the rule of both leaderboards we use), the *median*, and the *minimum* (worst-task score). Rank rules use only the order of models on each task: *Borda* (sum over tasks of the number of models beaten, ties counting one half; equivalent to mean rank) and the *Condorcet-winner* requirement (the target beats every rival on a strict majority of tasks, ties counting one half). Being a Condorcet winner is sufficient for being first under any Condorcet-consistent rule such as Copeland or Schulze, so its investment upper-bounds theirs. We also solve *plurality* (strictly the most first places, ties split) and *Copeland* (pairwise wins, ties one half) exactly; Copeland only on HELM, where its full pairwise model fits ($M = 79$). On HELM the Copeland optimum equals the Condorcet-requirement optimum on every target (same 25 of 78 feasible targets under the primary cap, $L_1$ equal to within $\varepsilon$), so the sufficient condition is tight there. Geometric, harmonic and power means and the trimmed mean are in the supplementary results only.

### 3.4 Exact minimum-investment solvers

For a score rule let $\tau = \max_{j \neq i} f(X_{j\cdot})$ be the level of the best rival; the target is first iff $f(y) \ge \tau + \varepsilon$. For the mean, the minimum and the median the minimum investment has a closed form (Propositions 1–3, proved in Appendix A); for power means and the trimmed mean we solve a convex program or enumerate exactly. For rank rules a raise of task $t$ to level $s + \varepsilon$ passes exactly the models with score at most $s$ on that task, so only these finitely many levels need be considered, and we solve the resulting integer program exactly with HiGHS, adding rival constraints lazily. Every solver is checked against brute-force search on small random instances (unit tests in the supplementary code). For each target we report $L_1$ and $K$, each minimised separately, and $L_\infty$ for the uniform baseline.

**Proposition 1** (Mean). *Let $f$ be the mean and $h_t = c_t - x_t$. The target can become first iff $\sum_t c_t > n\tau$ (up to $\varepsilon$). The minimum $L_1$ is $n(\tau - f(x))$ regardless of how it is spread, the minimum $K$ is the least $k$ for which the $k$ largest $h_t$ sum to at least this amount, and the minimum $L_\infty$ is $\tau - f(x)$ (raise all tasks equally) when no cap binds.*

**Proposition 2** (Minimum). *Let $f = \min_t y_t$. The target can become first iff $c_t > \tau$ for all $t$ with $x_t \le \tau$. Then $L_1 = \sum_t (\tau - x_t)^+$ and $K = \#\{t : x_t \le \tau\}$: every deficient task must be raised.*

**Proposition 3** (Median, $n$ odd). *Let $m = (n+1)/2$, $d_t = (\tau - x_t)^+$ for tasks with $c_t > \tau$ and $d_t = \infty$ otherwise. The target is first iff at least $m$ tasks end above $\tau$. The minimum $L_1$ is the sum of the $(m - \#\{t : x_t > \tau\})^+$ smallest finite $d_t$ among tasks with $x_t \le \tau$, and $K$ equals that count; the target can be made first iff enough finite $d_t$ exist.*

(All statements are for $\varepsilon \to 0$; the code uses $\varepsilon = 10^{-3}$ points. For even $n$ the median is the average of the two middle order statistics and we only have the computed solution.) We verify Propositions 1–3 numerically against the solvers (Section 4); no theorem is claimed for the rank rules.

# 4 Experimental setup

**Data.** We use two public leaderboards, each at a fixed, recorded version (checksums in the supplementary reproduction notes). *HELM MMLU*, release v1.13.0 (summary dated 2025-01-10): the exact-match score of 79 models on each of the 57 MMLU subjects ($79 \times 57$, no missing cell). *Open LLM Leaderboard* (OLLB) `contents`, dataset commit `9c09a7c` (2025-03-20): 4576 rows, of which 4497 have distinct model names after we keep the best-scoring variant of each duplicated name, and 4496 after removing one flagged model; each model has six normalised task scores (IFEval, BBH, MATH Lvl 5, GPQA, MUSR, MMLU-PRO), so $n = 6$. Merged models are kept in the main analysis and dropped in a robustness slice. The mean we use equals the sites' own aggregate (OLLB `Average`, HELM "MMLU All Subjects") to within $6 \times 10^{-14}$ points on every model. Neither source declares a licence for the score data; we redistribute only the derived score tables.

**Targets, caps and what is reported.** Every model other than the current leader is a target in turn, with all others fixed (4495 targets on OLLB, 78 on HELM). On OLLB the primary cap is `emp99` (per-task 99th percentile of the gain of a fine-tune over its declared base model, estimated from 540 such pairs in the same leaderboard). HELM has no base/fine-tune pairs, so its primary cap, `emp99T`, applies the OLLB MMLU-PRO 99th-percentile gain to all 57 subjects; this is a transfer assumption, not a measurement on HELM. The oracle cap `ceil` is reported beside it. All investment statistics are medians over feasible targets, shown together with the feasible fraction. OLLB rank rules are exact integer programs, which at $M = 4496$ models we run for the 50 best targets under the mean, and for all 499 targets of the slice made of the 500 best models; we apply a 120 s limit per instance and report time-outs as censored, never as infeasible (one Borda instance on the slice).

# 5 Results

All statements below are about targets that can be made first within the cap; the feasible fraction is shown beside every cost, and a rule is called cheaper than another only on targets that are feasible under both.

**The closed forms are exact.** Propositions 1–3 agree with the repository's own exact score-rule solvers on every target of both datasets and every cap, and with an independent generic LP/MILP formulation and brute-force enumeration on a 60-target sample and on small random instances (unit tests): the largest deviation in $L_1$ is below $2 \times 10^{-12}$ points, the deviation in $K$ is zero, and no feasibility verdict differs (Appendix Table 3). The OLLB median is solver-only because $n = 6$ is even.

**Feasibility depends on rank and on the rule.** Under the data-grounded caps, targeted training cannot make most models first (Figure 1, left). On OLLB the feasible fraction under the mean and the median is close to one for targets up to rank a few hundred and near zero beyond rank about one thousand; on HELM every target at ranks 2–10 is feasible under the mean, median, Borda and Condorcet rules, and the fraction falls steeply afterwards (Table 2: for ranks 11–50 only 6 of 40 targets are feasible under the mean and 17 of 40 under the median). The minimum rule is the most restrictive on both datasets, since every task below the best rival's weakest score must be raised and any one cap can block it (Proposition 2). The cap, not the rule, decides feasibility far from the top: with the oracle cap `ceil` every target is feasible (Appendix Figure 3).

Table 1 gives the frontier. On OLLB the fraction of targets that can reach first place under `emp99` is 12% under the mean, 10% under the median and 6% under the minimum; every target is feasible only up to rank 394 (mean), 90 (median) and 5 (minimum), and the largest feasible rank is 901, 1207 and 2225. On HELM under `emp99T` the fractions are 19%, 33% and 12%, every target is feasible up to rank 16, 19 and 6, and the largest feasible rank is 16, 29 and 20; the Condorcet requirement reaches rank 26 and Borda 17. On HELM the median is thus more exposed to targeted training than the mean on all three measures. On OLLB the shares are close and the intervals nearly touch (0.10 against

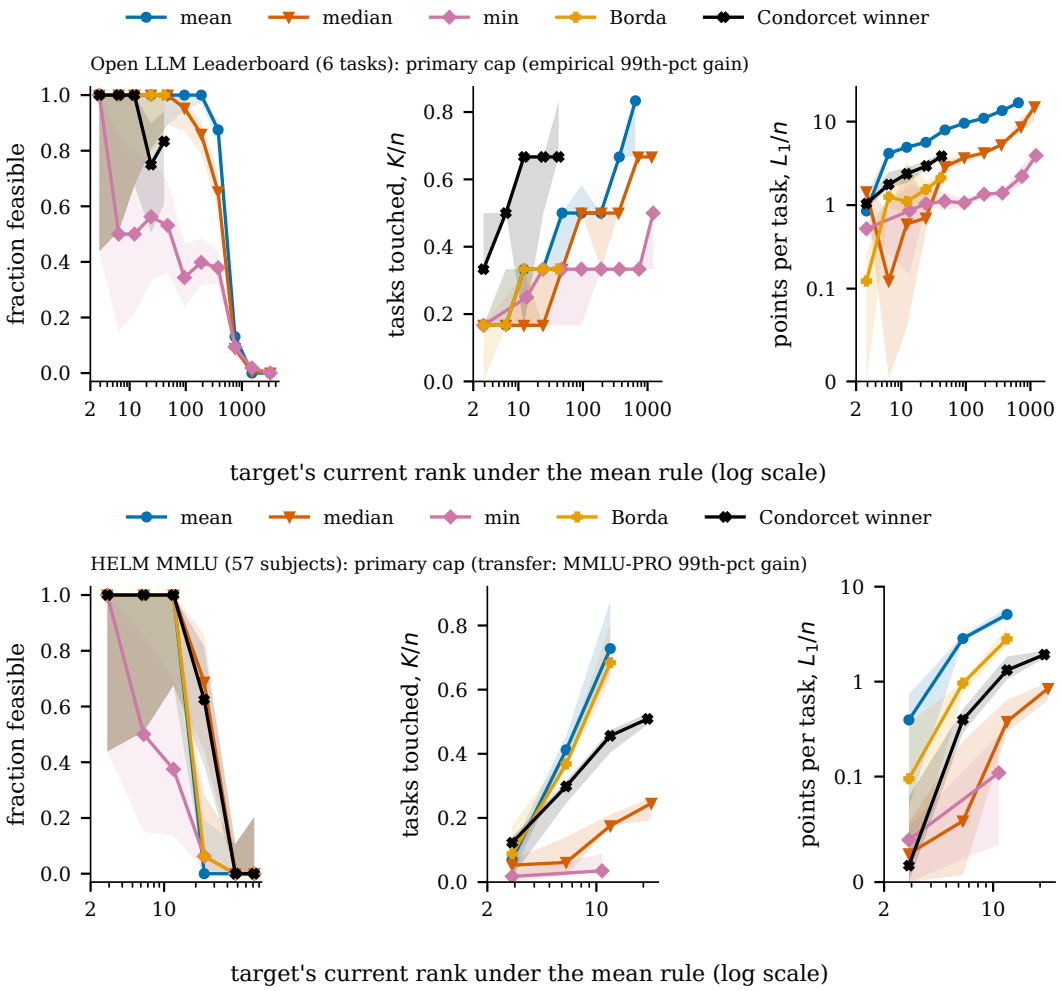



Figure 1: Minimum targeted investment to become first versus the target's current rank under the mean, primary caps. Left: fraction of targets that can reach first at all. Middle and right: tasks touched ($K/n$) and total points per task ($L_1/n$), *among feasible targets*. Top: OLLB, $n = 6$, cap `emp99` (4495 targets; Borda and Condorcet only for the 50 best). Bottom: HELM MMLU, $n = 57$, cap `emp99T` (78 targets; the cap is transferred from OLLB). Points are bin medians (bins double from rank 2; bins with fewer than 3 targets omitted), bands are 95% bootstrap intervals over targets. Generated by `figures/src/fig_rank_rules.py`.

0.12): the mean is feasible for more targets and for a longer unbroken run of ranks, the median has the single largest feasible rank (which may be an isolated case), so no ordering of exposure between the mean and the median is established there. The minimum is feasible rarely on both datasets but, when feasible, cheaply. The largest feasible rank can be an isolated case (OLLB, minimum), which is why the column "all feasible up to" is shown. Plurality is the most exposed rule on HELM: under `emp99T` 45 of 78 targets can win (58%), every target up to rank 39 and one as low as rank 59. The ordering of feasible shares on HELM does not depend on the cap: for caps of +5, +10 and +20 points per task and for `emp99T`, plurality exceeds the median, which is at or above Condorcet and Copeland, which exceed Borda and the mean (these two swap places between caps), which exceed the minimum (at +10 points: 73%, 51%, 49%, 32%, 37% and 18%; Appendix Figure 5 and Table 4).

**Cost, conditional on feasibility, differs strongly by rule.** Among feasible targets, the mean rule is the most expensive on both datasets and the minimum and median rules the cheapest (Figure 1, right; Table 2). On OLLB at ranks 11–50 the median feasible target needs 37.8 points in total under the mean, 9.1 under the median, 6.9 under the minimum, 11.7 under Borda and 20.5 under the

Table 1: Feasibility frontier under the primary cap (`emp99` on OLLB, `emp99T` on HELM) and the oracle cap (`ceil`). *share*: fraction of targets for which first place is reachable (Wilson 95% intervals); $r^*$: largest rank under the mean among feasible targets (may be an isolated case); *all feas. up to*: largest $r$ such that every target of rank $\leq r$ is feasible; *first bin < 1/2*: first rank bin (bins double from rank 2) in which fewer than half are feasible. †: the 50 best targets under the mean only, not comparable with the score rules' fractions. Generated by `figures/src/tables.py`.

| cap | rule | targets | feasible | share [95% CI] | $r^*$ | all feas. up to | first bin < 1/2 |
|---|---|---|---|---|---|---|---|
| primary (`emp99`) | mean | 4495 | 546 | 0.12 [0.11, 0.13] | 901 | 394 | 513–1024 |
| primary (`emp99`) | median | 4495 | 451 | 0.10 [0.09, 0.11] | 1207 | 90 | 513–1024 |
| primary (`emp99`) | min | 4495 | 273 | 0.06 [0.05, 0.07] | 2225 | 5 | 65–128 |
| primary (`emp99`) | Borda† | 49 | 49 | 1.00 [0.93, 1.00] | 50 | 50 | none |
| primary (`emp99`) | Condorcet† | 49 | 42 | 0.86 [0.73, 0.93] | 50 | 18 | none |
| oracle (`ceil`) | mean | 4495 | 4495 | 1.00 [1.00, 1.00] | 4496 | 4496 | none |
| oracle (`ceil`) | median | 4495 | 4495 | 1.00 [1.00, 1.00] | 4496 | 4496 | none |
| oracle (`ceil`) | min | 4495 | 4495 | 1.00 [1.00, 1.00] | 4496 | 4496 | none |
| oracle (`ceil`) | Borda† | 49 | 49 | 1.00 [0.93, 1.00] | 50 | 50 | none |
| oracle (`ceil`) | Condorcet† | 49 | 49 | 1.00 [0.93, 1.00] | 50 | 50 | none |

| cap | rule | targets | feasible | share [95% CI] | $r^*$ | all feas. up to | first bin < 1/2 |
|---|---|---|---|---|---|---|---|
| primary (`emp99T`) | mean | 78 | 15 | 0.19 [0.12, 0.29] | 16 | 16 | 17–32 |
| primary (`emp99T`) | median | 78 | 26 | 0.33 [0.24, 0.44] | 29 | 19 | 33–64 |
| primary (`emp99T`) | min | 78 | 9 | 0.12 [0.06, 0.20] | 20 | 6 | 9–16 |
| primary (`emp99T`) | Borda | 78 | 16 | 0.21 [0.13, 0.31] | 17 | 17 | 17–32 |
| primary (`emp99T`) | Condorcet | 78 | 25 | 0.32 [0.23, 0.43] | 26 | 26 | 33–64 |
| oracle (`ceil`) | mean | 78 | 78 | 1.00 [0.95, 1.00] | 79 | 79 | none |
| oracle (`ceil`) | median | 78 | 78 | 1.00 [0.95, 1.00] | 79 | 79 | none |
| oracle (`ceil`) | min | 78 | 78 | 1.00 [0.95, 1.00] | 79 | 79 | none |
| oracle (`ceil`) | Borda | 78 | 78 | 1.00 [0.95, 1.00] | 79 | 79 | none |
| oracle (`ceil`) | Condorcet | 78 | 78 | 1.00 [0.95, 1.00] | 79 | 79 | none |

Table 2: Median investment to become first among feasible targets, by rank stratum and rule, primary caps: OLLB (`emp99`, top) and HELM (`emp99T`, bottom). "feas." = feasible targets / targets in the stratum; $K$, $L_1$ with 95% bootstrap intervals over targets; the last column is the median ratio of $L_1$ to the mean rule over targets feasible under both. OLLB Borda and Condorcet use the 50 best targets only. Generated by `figures/src/tables.py`.

| | rank 2–10 | | | | rank 11–50 | | | |
|---|---|---|---|---|---|---|---|---|
| | feas. | $K$ | $L_1$ | $L_1/L_1^{\text{mean}}$ | feas. | $K$ | $L_1$ | $L_1/L_1^{\text{mean}}$ |
| mean | 9/9 | 1.0 [1.0, 2.0] | 24.6 [5.1, 27.7] | – | 40/40 | 2.0 [2.0, 2.0] | 37.8 [33.9, 41.0] | – |
| median | 9/9 | 1.0 [1.0, 1.0] | 0.8 [0.0, 8.4] | 0.03 [0.00, 1.70] (9) | 40/40 | 1.0 [1.0, 2.0] | 9.1 [4.4, 14.6] | 0.22 [0.13, 0.40] (40) |
| min | 5/9 | 1.0 [1.0, 2.0] | 3.7 [2.8, 14.6] | 0.61 [0.23, 0.77] (5) | 24/40 | 2.0 [1.5, 2.0] | 6.9 [5.9, 8.4] | 0.18 [0.14, 0.19] (24) |
| Borda | 9/9 | 1.0 [1.0, 2.0] | 5.6 [0.7, 6.2] | 0.20 [0.09, 0.35] (9) | 40/40 | 2.0 [2.0, 2.0] | 11.7 [9.5, 12.6] | 0.27 [0.25, 0.31] (40) |
| Condorcet | 9/9 | 3.0 [2.0, 4.0] | 10.0 [6.7, 16.7] | 0.59 [0.41, 1.04] (9) | 33/40 | 4.0 [4.0, 4.0] | 20.5 [18.0, 22.4] | 0.50 [0.47, 0.55] (33) |

| | rank 2–10 | | | | rank 11–50 | | | |
|---|---|---|---|---|---|---|---|---|
| | feas. | $K$ | $L_1$ | $L_1/L_1^{\text{mean}}$ | feas. | $K$ | $L_1$ | $L_1/L_1^{\text{mean}}$ |
| mean | 9/9 | 23.0 [4.0, 32.0] | 156.3 [22.5, 222.4] | – | 6/40 | 46.0 [40.0, 51.0] | 321.2 [278.2, 351.3] | – |
| median | 9/9 | 4.0 [1.0, 10.0] | 3.3 [0.4, 16.6] | 0.04 [0.00, 0.09] (9) | 17/40 | 12.0 [10.0, 15.0] | 37.5 [31.7, 53.6] | 0.09 [0.07, 0.11] (6) |
| min | 6/9 | 1.5 [0.5, 2.5] | 2.3 [1.0, 5.3] | 0.03 [0.00, 0.09] (6) | 3/40 | 3.0 [2.0, 5.0] | 10.4 [6.2, 15.5] | 0.04 [0.02, 0.06] (2) |
| Borda | 9/9 | 20.0 [5.0, 30.0] | 54.0 [5.6, 100.5] | 0.36 [0.28, 0.94] (9) | 7/40 | 45.0 [38.0, 48.0] | 187.8 [143.1, 230.8] | 0.55 [0.51, 0.64] (6) |
| Condorcet | 9/9 | 16.0 [7.0, 22.0] | 20.7 [0.9, 57.6] | 0.16 [0.02, 0.22] (9) | 16/40 | 28.0 [27.0, 29.0] | 104.6 [90.4, 113.9] | 0.28 [0.21, 0.31] (6) |

Condorcet-winner requirement; relative to the mean on jointly feasible targets, the ratios are 0.22, 0.18, 0.27 and 0.50. On HELM at ranks 11–50 the corresponding totals are 321, 38, 10, 188 and 105 points. The reason is structural: the mean must close the whole summed gap ($n$ times the gap, Proposition 1), whereas order-statistic rules pay only for the cheapest majority of tasks or the weakest tasks. Rank rules sit in between, because they pay for passing rivals task by task. Table 2 shows intervals; the ordering mean > Condorcet > Borda $\gtrsim$ median $\gtrsim$ min on OLLB is not uniform across rank bins (the intervals at ranks 2–10 are wide, with $n = 9$ targets).

**Rank rules on a larger OLLB slice.** Restricting OLLB to its 500 best models lets us solve every rule exactly for all 499 targets (cap `emp99`; the slice is a smaller leaderboard, its shares are not those

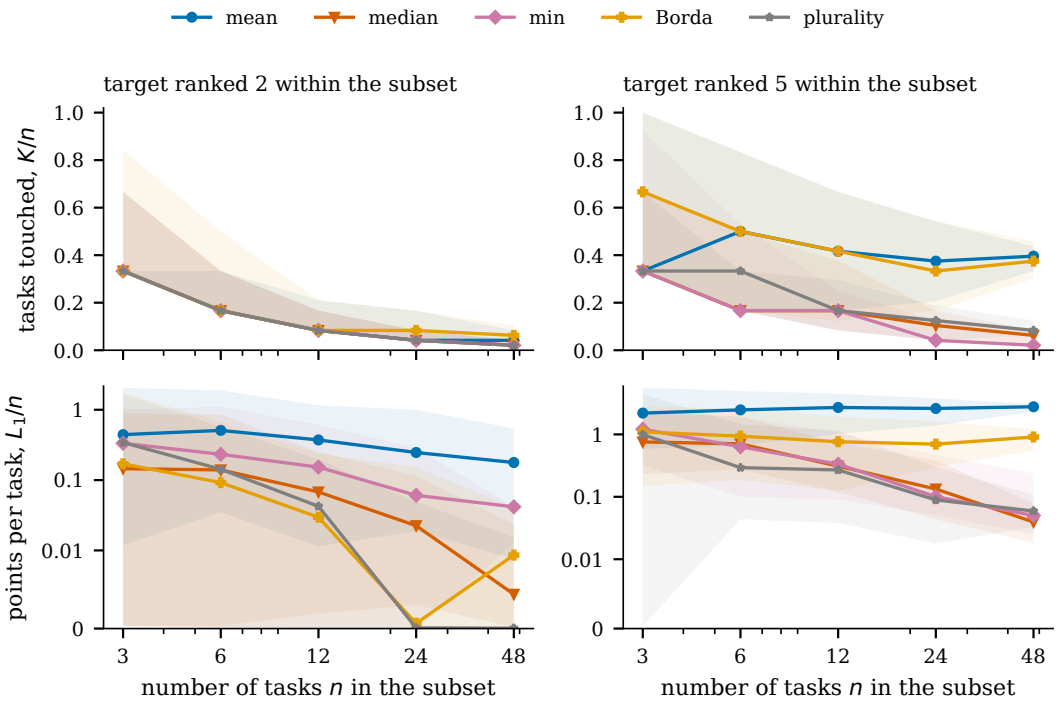



Figure 2: HELM: investment versus the number of tasks $n$ in a random subset of subjects, for the target ranked second (left) or fifth (right) within the subset under the mean; bands span the 2.5–97.5 percentiles over 60 subset draws, among feasible draws, cap `emp99T`. Generated by `figures/src/fig_scaling.py`.

of the full board). The feasible shares are 0.95 for the mean, 0.91 for Borda, 0.79 for the median, 0.56 for plurality, 0.40 for the minimum and 0.30 for the Condorcet requirement; the median total gains among feasible targets are 70.3, 36.5, 25.5, 29.3, 7.6 and 42.2 points. Cost thus follows mean $>$ Condorcet $>$ Borda $>$ plurality $\gtrsim$ median $>$ minimum, as in the top-50 results, while feasibility of the Condorcet requirement is far below that of the mean on OLLB ($n = 6$, four tasks must be won) and above it on HELM ($n = 57$). One Borda instance timed out and is censored, not counted as infeasible.

**Which of our predictions held.** The plan fixed three hypotheses before the full results (H1–H3, Appendix D). H2 (cost rises with rank, feasibility falls as caps tighten) holds on both datasets. H1 (order-statistic and majority rules need *more* tasks but possibly fewer points than the mean) holds only in part: they need fewer points, but not more tasks, except that the Condorcet requirement touches more tasks than the mean on OLLB at $n = 6$ ($K = 4$ against 2 at ranks 11–50); on HELM the median needs fewer tasks than the mean ($K = 12$ against 46). H3 is examined next.

**Number of tasks.** Figure 2 draws random subsets of $n \in \{3, 6, 12, 24, 48\}$ HELM subjects and takes the model ranked second or fifth within each subset as the target. For the rank-2 target $K/n$ falls with $n$ under every rule, and for the rank-5 target the mean and Borda rules keep touching about 0.4 of the tasks while the median, minimum and plurality fall. For the rank-5 target the mean rule's $L_1/n$ stays roughly constant in $n$ while that of the median, minimum and plurality falls; for the rank-2 target all rules fall. H3 (that $K/n$ of the median and majority rules does not shrink) is therefore not supported for the median. The bands are wide (60 subset draws), so we make no claim about differences between neighbouring $n$. OLLB, with $n = 6$, is a single small-$n$ point and supports no scaling claim.

**Robustness to noise and task choice.** Appendix Figure 6 resamples the leaderboard for target models at fixed ranks (OLLB ranks 2–100, HELM ranks 2–40) in two ways: evaluation noise (binomial standard errors from item counts; 300 draws for the OLLB closed-form rules, 60 for OLLB

Borda, Condorcet and plurality, 200 for all HELM rules) and a bootstrap over tasks (same caps, same draw counts). Wherever a target can be made first on the observed data, the observed $L_1$ lies inside the 2.5–97.5% interval of the evaluation-noise draws, in all 66 cases (24 OLLB closed-form, 16 OLLB ordinal, 26 HELM); it is below the draw median in 55 of them, by a few percent in the examples we checked (28.5 against 29.9 points for the OLLB rank-10 target under the mean; 154 against 162 for the HELM rank-5 target). The intervals are wide: for the HELM rank-5 target the mean rule's interval is 114–226 points. Feasibility near the frontier is noise-sensitive. The HELM rank-20 target, infeasible under the mean on the observed data, is feasible in 4.5% of the noise draws (median cost 385 points) and the rank-40 target in none; under OLLB plurality (not drawn in the figure; `results/summary/summary_rob_ollb_noise_ord_emp99.csv`) the rank-20 and rank-50 targets are infeasible on the observed data and feasible in 2% and 38% of the draws; for the OLLB minimum rule most draws are infeasible for most targets. A target whose gap to first place is within the evaluation noise therefore cannot be said to need any particular investment: a win by the margin $\varepsilon$ would not survive resampling.

**Sensitivity to margin, slice, scale, weights and cost unit.** Appendix Tables 5–7 report the variants. A margin of 1 or 2 points instead of $10^{-3}$ lowers the feasible share and raises the cost: on OLLB under the mean the share goes from 12.1% to 11.1% and 10.4% (cost ratio 1.08 and 1.17), for the minimum from 6.1% to 4.3% and 2.6%; on HELM the mean falls from 19% to 15% and 12% (ratio 1.33 and 1.73) and the median from 33% to 28% and 21% (ratio 1.55 and 2.34). The rule ordering in cost is unchanged on HELM (at 2 points: mean 270, Borda 117, Condorcet 90, plurality 43, median 39, minimum 6.7 points). Removing merged models (3788 targets) leaves the mean and median costs unchanged (ratio 1.00) and the minimum rule feasible for 7.0% of targets instead of 5.6% (cost ratio 0.81). With raw, un-normalised task accuracies and the oracle cap the order mean $>$ median $>$ minimum stays (154, 93, 75 points) but the gap between median and minimum narrows. Measuring cost in standard deviations of the task across models preserves the order on both datasets under both caps. Task weights matter as much as the rule: with Dirichlet(1) weights the cost of the same target is 0.54 (OLLB) and 0.43 (HELM) of its cost under equal weights in the median over targets feasible under both, and 0.70 and 0.68 with Dirichlet(5) weights; the leaderboards' own equal weights are one point of this family, and the target's rank under the weights differs from its rank under the mean.

## 6 Discussion and limitations

**What the results say.** Under caps taken from observed fine-tuning gains, a developer who can lift only the target's own scores, and whose rivals stand still, tops these leaderboards only from near the top, and how near depends on the rule: on HELM the median reaches further down than the mean, on OLLB the two are not ordered (the mean is feasible for more targets and a longer unbroken run of ranks, the median has the largest feasible rank), and the minimum is rarely feasible on both. Where it is feasible the mean rule demands the most points, because it must close $n$ times the gap (Proposition 1) whatever the spread, while the median and minimum pay only for a majority of tasks or for the weak tasks. Among the score rules the mean is thus the costliest to manipulate in points, among feasible targets; in the rank that can be reached it is harder to move than the median on HELM and not established to differ from it on OLLB, while the minimum rule is the most restrictive on both. The median is cheap in points (and reaches further on HELM), and the minimum is cheap when feasible but needs every weak task to be raised and so is limited by the cap. The rank rules lie between the mean and the median in points, but their mutual order is not stable: Condorcet is dearer than Borda on OLLB ($n = 6$) and cheaper on HELM ($n = 57$), and its feasibility is far below the mean's on the OLLB top-500 slice and above it on HELM. On HELM Borda reaches only one rank further than the mean (rank 17 against 16), whereas plurality, which needs only the most first places, reaches rank 59. Weights are a further lever of the same size as the rule: random task weights change the cost of a target by 30–57% in the median. These orderings come with caveats we state next.

**Conditions under which the conclusions fail or are not established.** (1) *The cap.* Feasibility is determined by the cap, not by the rule, far from the top (oracle rows of Table 1). The OLLB cap uses gains of published fine-tunes over their declared base models; this is selection-biased toward successes and says nothing about what a determined developer could achieve with a new data source. The HELM cap is a transfer of the OLLB MMLU-PRO gain to 57 subjects and has not been measured on HELM. Headline claims are therefore stated per dataset. (2) *Independence across tasks.* We let

each task be raised at no cost to others; targeted training can trade off tasks, which would make the true cost higher. (3) *Fixed rivals.* Other developers do not respond; leaderboards with many strategic developers need a game model such as that of Chen et al. [2026]. (4) *Score tables, not training.* The investment is in score points, not compute, data or money, so "cheap" means few score points, not little effort. (5) *Ranks within noise.* The margin of $10^{-3}$ points that defines "first" is below the evaluation noise of the scores; a win by that margin would not survive resampling (Figure 6), and the intervals for the median and minimum are wide. (6) *Rule coverage and size.* On the full OLLB board rank rules were solved exactly for the 50 best targets only (and for all 499 targets of the top-500 slice), so their feasibility fractions cannot be compared with those of the score rules; the Condorcet requirement is a sufficient condition for Condorcet-consistent rules and is tight for Copeland on HELM, but Copeland was not solved on OLLB; Kemeny and Schulze were not studied. (7) *Small $n$ and small $M$.* OLLB has six tasks and HELM 78 targets, so rank bins on HELM hold few targets and their intervals are wide; scaling in $n$ comes from HELM subsets only and is a statement about random subsets of one benchmark family. (8) *Data snapshot.* Both leaderboards are frozen snapshots (OLLB 2025-03, HELM v1.13.0); we do not claim anything about later models. (9) *Unfinished and partial items.* The OLLB ordinal-rule noise and bootstrap runs use 60 draws and cover seven target ranks, not all 50 targets; Copeland on the OLLB top-500 slice was not delivered; one Borda instance on that slice is censored. The OLLB evaluation-noise model takes item counts from one results file and assumes them equal for all models. (10) *Weights and cost unit.* All rules use equal task weights and points as the cost unit, as the leaderboards do; the weight and standard-deviation variants show the cost depends on both.

**Broader impact.** The analysis tells leaderboard designers how exposed each rule is to targeted training under stated assumptions, which supports choosing rules and caps (for instance holding out sub-datasets) deliberately. It could also be read as a recipe for gaming a ranking. We think the risk is small: it needs no new capability beyond fine-tuning on data a developer already can use, and our results indicate that for most models it is infeasible under the observed gains. We do not evaluate any model's training.

## 7 Reproducibility

All numbers, tables and figures are produced by the supplementary code from the pinned public data (versions, checksums, seeds, commands, expected run times and a table mapping every figure and table to its generating script are in `REPRODUCE.md`). The solvers are deterministic; random draws use seed 0. The full pipeline needs no GPU; the wall-clock times of each step, taken from the run logs and listed in `REPRODUCE.md` (jobs shared the container, so they are not isolated timings), give roughly 8–10 hours run sequentially on two CPUs. The generators of every figure and table of this paper are listed in Appendix C; `code/build_figures.sh` regenerates them and builds the PDF (`cd paper && latexmk -pdf main.tex`). Neither leaderboard declares a licence for its score data; we release only derived score tables. The analysis recorded that no material about the published study that motivated this question was searched for or encountered (`REPRODUCE.md`, section on forbidden material).

## References


Felix Brandt, Vincent Conitzer, Ulle Endriss, Jérôme Lang, and Ariel D. Procaccia, editors. *Handbook of Computational Social Choice*. Cambridge University Press, 2016. doi: 10.1017/CBO9781107446984.

Yatong Chen, Guanhua Zhang, and Moritz Hardt. Leaderboard incentives: Model rankings under strategic post-training, 2026. arXiv:2603.08371.

Pierre Colombo, Nathan Noiry, Ekhine Irurozki, and Stephan Clemencon. What are the best systems? new perspectives on nlp benchmarking, 2022. arXiv:2202.03799.

Mostafa Dehghani, Yi Tay, Alexey A. Gritsenko, Zhe Zhao, Neil Houlsby, Fernando Diaz, Donald Metzler, and Oriol Vinyals. The benchmark lottery, 2021. arXiv:2107.07002.

Piotr Faliszewski, Edith Hemaspaandra, and Lane A. Hemaspaandra. How hard is bribery in elections? *Journal of Artificial Intelligence Research*, 35:485–532, 2009. doi: 10.1613/jair.2676.

Dan Hendrycks, Collin Burns, Steven Basart, Andy Zou, Mantas Mazeika, Dawn Song, and Jacob Steinhardt. Measuring massive multitask language understanding, 2020. arXiv:2009.03300.

Marc Lanctot, Kate Larson, Yoram Bachrach, Luke Marris, Zun Li, Avishkar Bhoopchand, Thomas Anthony, Brian Tanner, and Anna Koop. Evaluating agents using social choice theory, 2023. arXiv:2312.03121.

Percy Liang, Rishi Bommasani, Tony Lee, Dimitris Tsipras, Dilara Soylu, Michihiro Yasunaga, Yian Zhang, Deepak Narayanan, Yuhuai Wu, Ananya Kumar, Benjamin Newman, Binhang Yuan, Bobby Yan, Ce Zhang, Christian Cosgrove, Christopher D. Manning, Christopher Ré, Diana Acosta-Navas, Drew A. Hudson, Eric Zelikman, Esin Durmus, Faisal Ladhak, Frieda Rong, Hongyu Ren, Huaxiu Yao, Jue Wang, Keshav Santhanam, Laurel Orr, Lucia Zheng, Mert Yuksekgonul, Mirac Suzgun, Nathan Kim, Neel Guha, Niladri Chatterji, Omar Khattab, Peter Henderson, Qian Huang, Ryan Chi, Sang Michael Xie, Shibani Santurkar, Surya Ganguli, Tatsunori Hashimoto, Thomas Icard, Tianyi Zhang, Vishrav Chaudhary, William Wang, Xuechen Li, Yifan Mai, Yuhui Zhang, and Yuta Koreeda. Holistic evaluation of language models, 2022. arXiv:2211.09110.

Shivalika Singh, Yiyang Nan, Alex Wang, Daniel D'Souza, Sayash Kapoor, Ahmet Üstün, Sanmi Koyejo, Yuntian Deng, Shayne Longpre, Noah A. Smith, Beyza Ermis, Marzieh Fadaee, and Sara Hooker. The leaderboard illusion, 2025. arXiv:2504.20879.

Mirac Suzgun, Nathan Scales, Nathanael Schärli, Sebastian Gehrmann, Yi Tay, Hyung Won Chung, Aakanksha Chowdhery, Quoc V. Le, Ed H. Chi, Denny Zhou, and Jason Wei. Challenging big-bench tasks and whether chain-of-thought can solve them, 2022. arXiv:2210.09261.

Shuo Yang, Wei-Lin Chiang, Lianmin Zheng, Joseph E. Gonzalez, and Ion Stoica. Rethinking benchmark and contamination for language models with rephrased samples, 2023. arXiv:2311.04850.

Kun Zhou, Yutao Zhu, Zhipeng Chen, Wentong Chen, Wayne Xin Zhao, Xu Chen, Yankai Lin, Ji-Rong Wen, and Jiawei Han. Don't make your llm an evaluation benchmark cheater, 2023. arXiv:2311.01964.

# A Proofs of Propositions 1–3

**Proposition 1.** $f(y) = \frac{1}{n}\sum_t y_t \geq \tau$ iff $\sum_t (y_t - x_t) \geq n(\tau - f(x))$, and $\sum_t (y_t - x_t) \leq \sum_t h_t$ with each $0 \leq y_t - x_t \leq h_t$; so feasibility is $\sum_t h_t \geq n(\tau - f(x))$, i.e. $\sum_t c_t \geq n\tau$, and the minimum of the (linear) total is the requirement itself, attained by any feasible spread. $K$ is minimised by filling the largest headrooms first, since touching $k$ tasks can supply at most the sum of the $k$ largest $h_t$. The uniform raise by $\tau - f(x)$ attains $L_\infty$, and no spread has a smaller maximum because the average raise is $\tau - f(x)$.

**Proposition 2.** $\min_t y_t > \tau$ requires $y_t > \tau$ for every $t$, hence $c_t > \tau$ wherever $x_t \leq \tau$ (tasks with $x_t > \tau$ need no change). The cheapest $y_t$ on a deficient task is $\tau$ (up to $\varepsilon$), so $L_1 = \sum_t (\tau - x_t)^+$, and $K$ counts the deficient tasks. Both are forced.

**Proposition 3.** For odd $n$ the median exceeds $\tau$ iff at least $m$ coordinates exceed $\tau$. Tasks with $x_t > \tau$ already count. We need $m' = (m - \#\{x_t > \tau\})^+$ more tasks. Lifting task $t$ above $\tau$ costs at least $d_t$ (and is possible iff $c_t > \tau$); costs are separable, so the cheapest set is the $m'$ smallest finite $d_t$ and $K = m'$, which cannot be smaller because each lifted task adds one to the count.

# B Further results

# C Reproducing the figures and tables

All commands run from the repository root; `R` is `results/`. The analysis (`code/run_all.sh`) produces `R`; the generators below only display it (bin medians with bootstrap intervals over targets, Wilson intervals), and each output has a sidecar recording its arguments and the SHA-256 of its inputs (`figures/MANIFEST.md`). Still running when this version was written: Copeland on the OLLB top-500 slice (a line of `code/run_all.sh`; not used by any number in this paper). Numbers quoted in

Table 3: Closed forms (Propositions 1–3) against the exact solvers and an independent generic LP/MILP (60-target sample), over all caps. Generated by `figures/src/tables.py` from `results/theory_check.csv`.

| data | rule | reference | max$\lvert\Delta L_1\rvert$ | max$\lvert\Delta K\rvert$ | feas. mismatches | caps |
|---|---|---|---|---|---|---|
| OLLB | mean | exact solver | 5.7e-14 | 0 | 0 | 7 |
| OLLB | mean | generic LP/MILP | 8.5e-14 | 0 | 0 | 7 |
| OLLB | min | exact solver | 0.0e+00 | 0 | 0 | 7 |
| OLLB | min | generic LP/MILP | 4.3e-14 | 0 | 0 | 7 |
| OLLB | median | exact solver | solver only | – | – | 7 |
| HELM | mean | exact solver | 4.5e-13 | 0 | 0 | 8 |
| HELM | mean | generic LP/MILP | 1.8e-12 | 0 | 0 | 8 |
| HELM | min | exact solver | 0.0e+00 | 0 | 0 | 8 |
| HELM | min | generic LP/MILP | 1.3e-12 | 0 | 0 | 8 |
| HELM | median | exact solver | 5.7e-14 | 0 | 0 | 8 |
| HELM | median | generic LP/MILP | 1.5e-12 | 0 | 0 | 8 |

Table 4: HELM feasibility frontier under the cap sweep, nine rules. Cell: number of the 78 targets for which first place is reachable (in parentheses: the largest rank $r$ such that every target of rank $\leq r$ under the mean is feasible; 79 means all). Caps: $+h$ points per subject at most; emp95$^T$ and emp99$^T$ transfer the OLLB MMLU-PRO gain percentiles to all subjects (emp99$^T$ is the primary cap); pooled: pooled six-task OLLB 99th percentile; front: best rival score per subject; 100: ceiling. Copeland was not run at emp95$^T$ and pooled; elsewhere it coincides with Condorcet. The prefix is a minimum over targets, so one early infeasible target lowers it (minimum rule at emp99$^T$: 9 feasible, prefix 6). Generated by `figures/src/tables_ext.py` from `results/summary/frontier_helm.csv`.

| rule | +5 | emp95$^T$ | +10 | emp99$^T$ | +20 | pooled | front | 100 |
|---|---|---|---|---|---|---|---|---|
| mean | 11 (12) | 7 (8) | 29 (30) | 15 (16) | 56 (55) | 77 (78) | 78 (79) | 78 (79) |
| median | 16 (13) | 10 (11) | 40 (33) | 26 (19) | 66 (67) | 77 (78) | 78 (79) | 78 (79) |
| min | 7 (6) | 5 (4) | 14 (6) | 9 (6) | 34 (9) | 77 (20) | 78 (79) | 78 (79) |
| Borda | 11 (9) | 8 (9) | 25 (19) | 16 (17) | 49 (43) | 76 (77) | 78 (79) | 78 (79) |
| Condorcet | 16 (16) | 10 (11) | 38 (37) | 25 (26) | 65 (65) | 77 (78) | 78 (79) | 78 (79) |
| Copeland | 16 (16) | – | 38 (37) | 25 (26) | 65 (65) | – | 78 (79) | 78 (79) |
| plurality | 29 (29) | 16 (12) | 57 (50) | 45 (39) | 74 (75) | 77 (78) | 78 (79) | 78 (79) |
| geometric mean | 8 (9) | 5 (6) | 24 (20) | 15 (16) | 50 (51) | 77 (78) | 78 (79) | 78 (79) |
| harmonic mean | 8 (9) | 4 (4) | 20 (20) | 13 (14) | 50 (51) | 77 (78) | 78 (79) | 78 (79) |

the text come from `R/summary/` (`frontier_*`, `paired_*`, `summary_rob_*`, `summary_variants`) as produced by `code/make_tables.sh`.

# D Hypotheses fixed before the full results

The plan fixed three hypotheses from the closed forms before the full result set was available. H1: with loose caps, score-sum rules need the same $L_1$ regardless of $K$, while order-statistic and majority rules need more tasks but possibly fewer points than the mean. H2: required investment grows with the target's initial rank; feasibility falls as caps tighten; rule ordering can change with the cap. H3: for fixed per-task caps the number of tasks that majority and median rules need does not shrink with $n$, while the mean rule's $L_1/n$ depends on the density of gaps. Anything beyond these is exploratory.

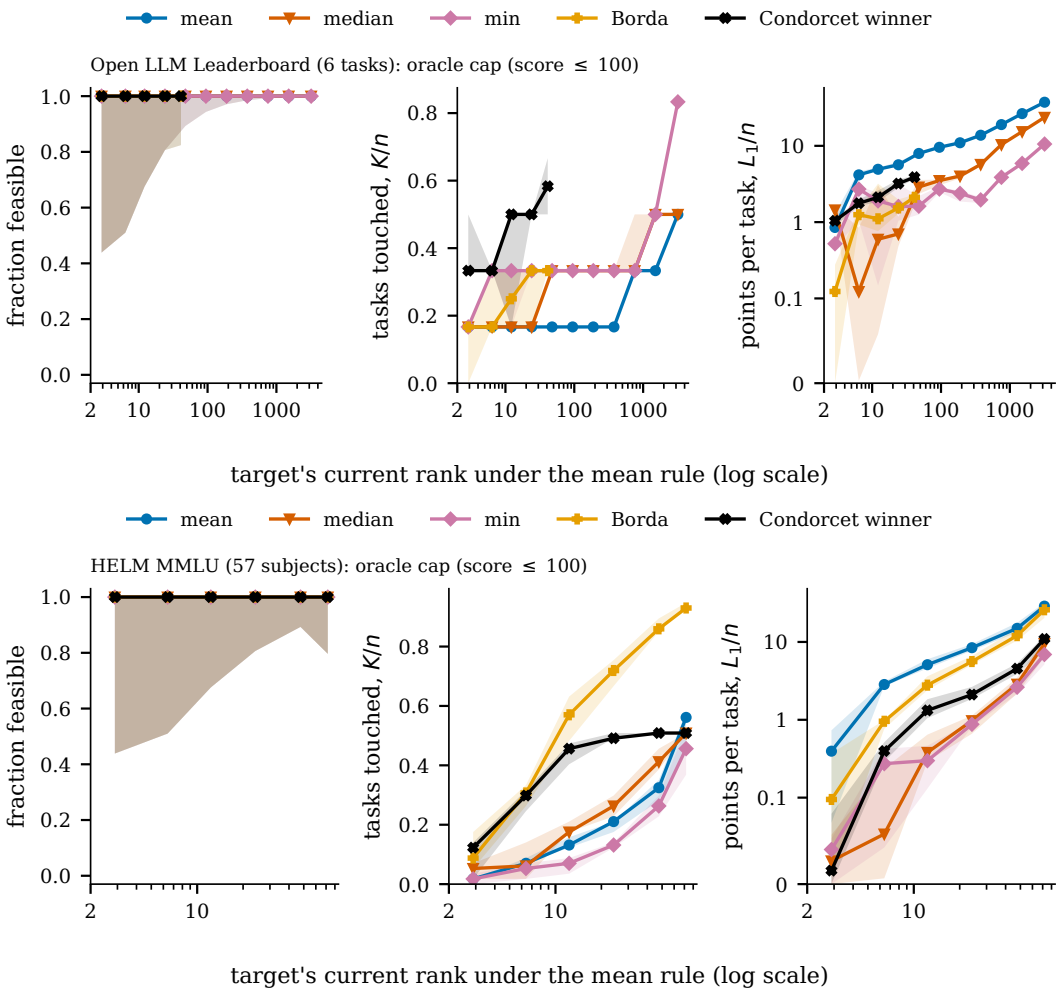



Figure 3: As Figure 1 with the oracle cap `ceil` (scores up to 100): a lower bound on any required investment; every target is feasible, so the differences between rules are not caused by the cap.

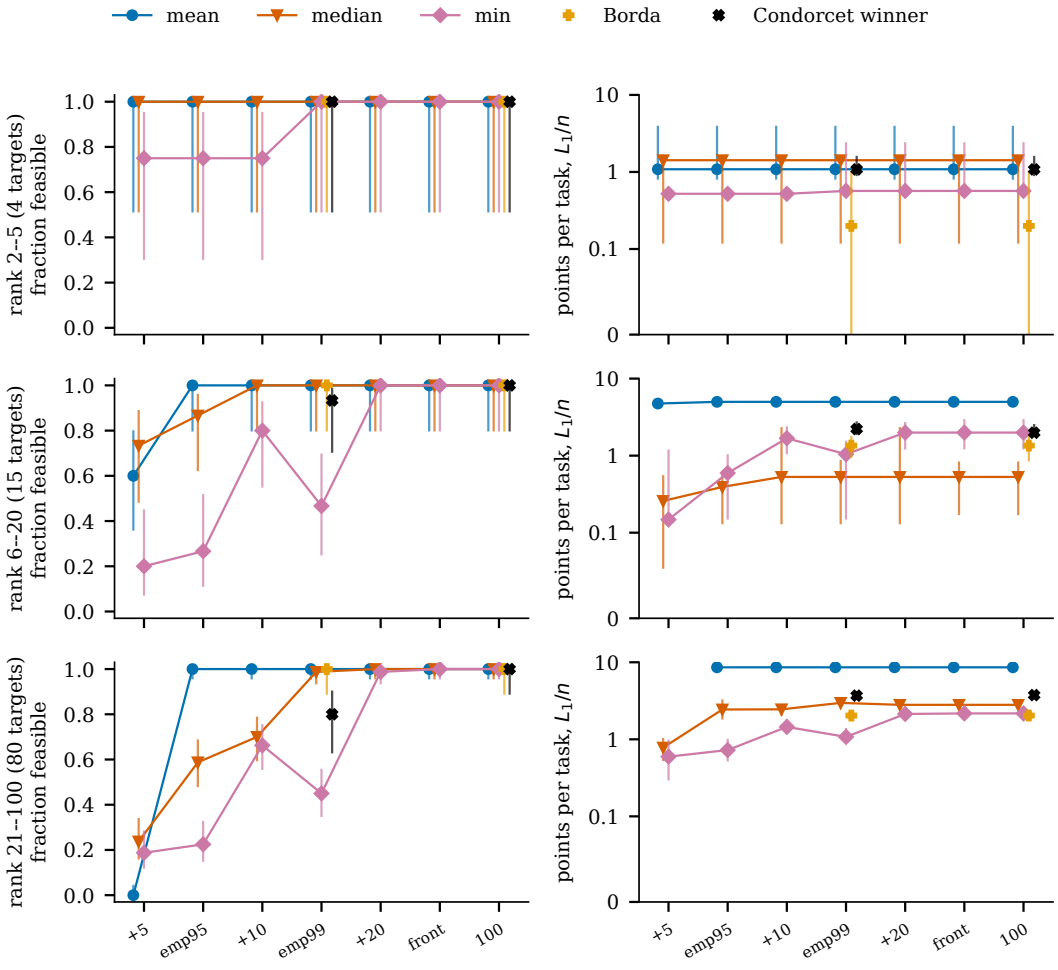
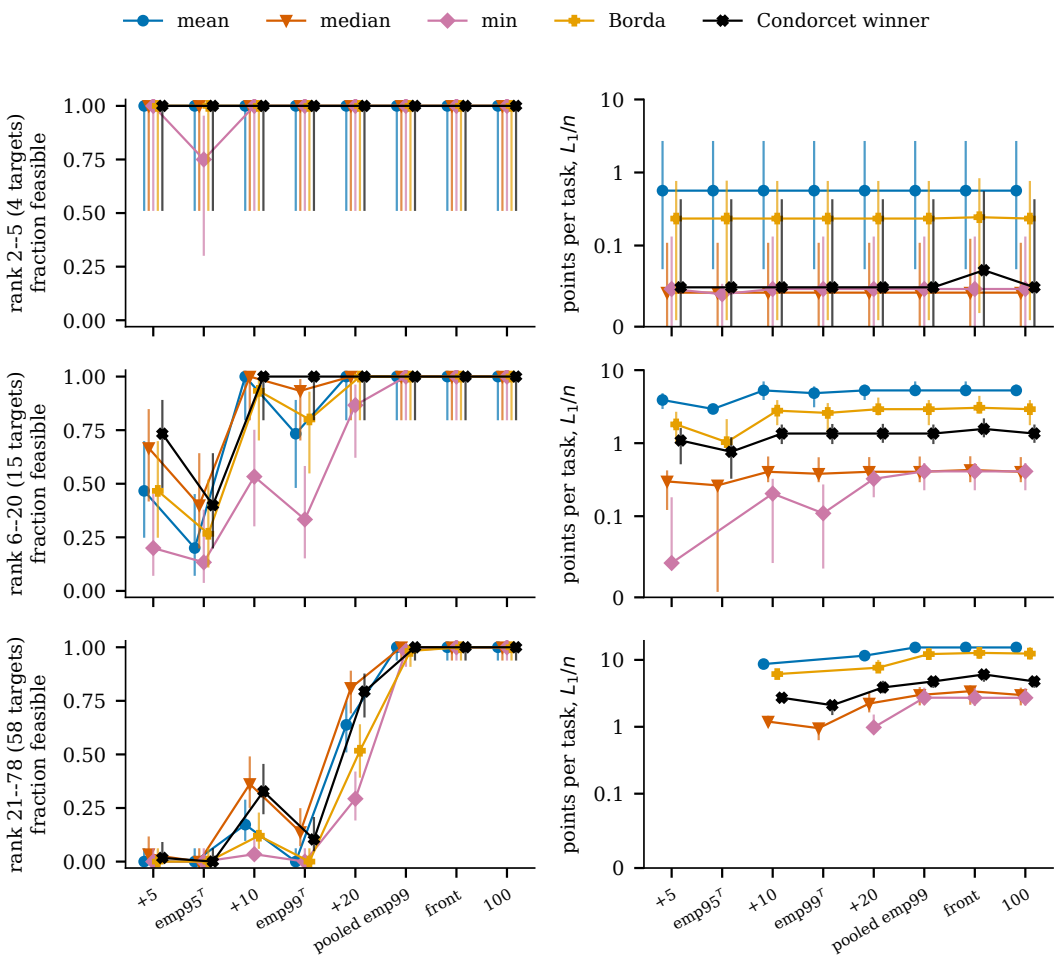



Figure 4: OLLB: effect of the cap on feasibility (left) and cost among feasible targets (right) for three rank strata. Caps run from tight to loose (+5 points per task, `emp95`, +10, `emp99`, +20, best rival score, 100). For the mean rule the cost among feasible targets does not depend on the cap (Proposition 1); only the feasible fraction does. Generated by `figures/src/fig_cap_grid.py`.




per-task cap on the gain, roughly tight → loose (+h: at most h points; emp: empirical fine-tune gain percentile; $^T$: transferred from OLLB; front: best rival score; 100: score ceiling)

Figure 5: HELM: effect of the cap on feasibility (left) and cost among feasible targets (right) for three rank strata, as Figure 4. Caps `emp95T` and `emp99T` transfer the OLLB MMLU-PRO gain percentiles to all 57 subjects. Generated by `figures/src/fig_cap_grid.py`.

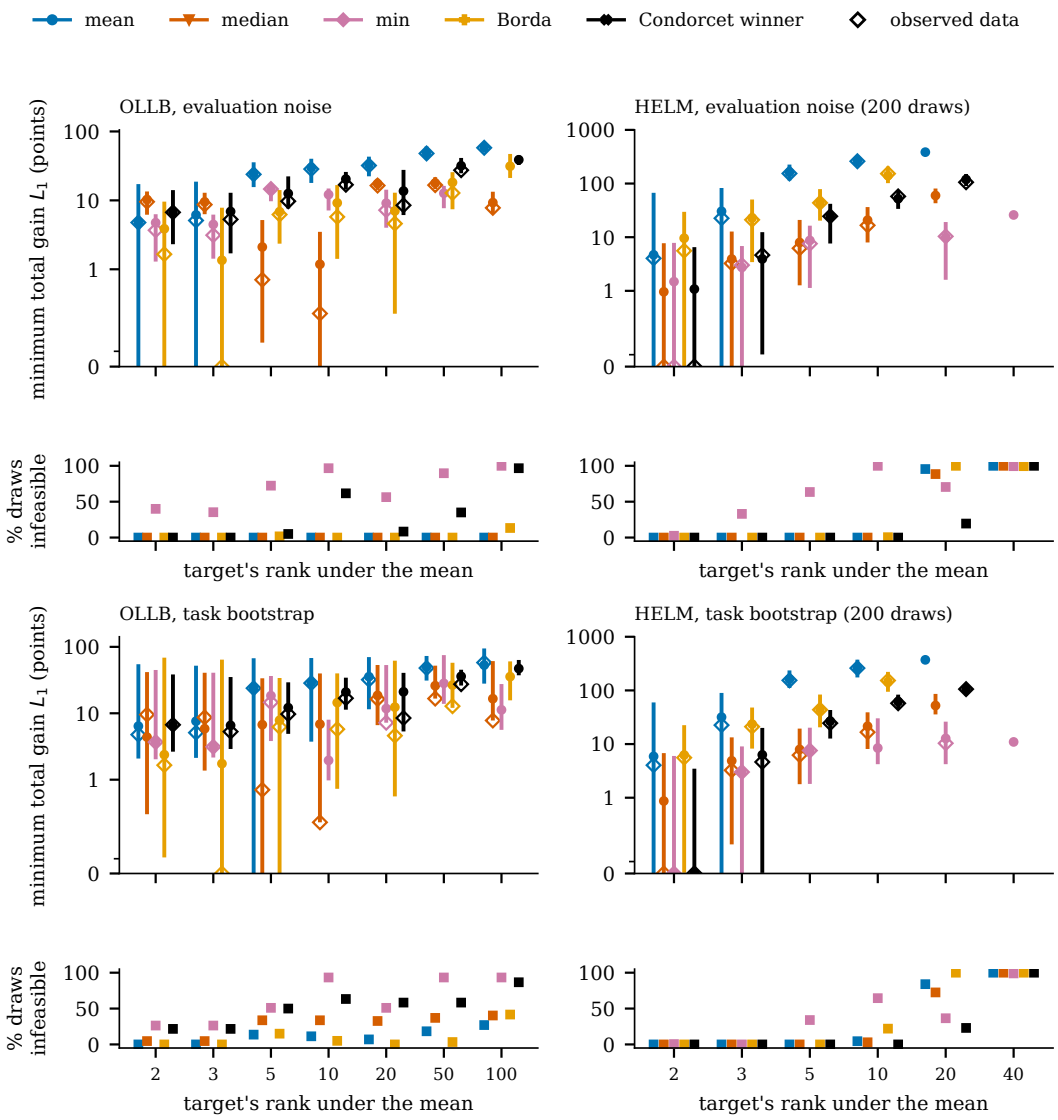



Figure 6: Stability of the minimum total gain $L_1$ and share of resampled leaderboards in which the target cannot become first, under evaluation noise and a task bootstrap, for OLLB and HELM, five rules (mean, median, minimum, Borda, Condorcet), primary caps. Dot: median over feasible draws; bar: 2.5–97.5 percentile of draws (a spread of draws, not an interval for the median); open diamond: value on the observed data; lower strip: percentage of draws in which first place is unreachable. OLLB Borda and Condorcet cover the strata targets within the top 50 only. Generated by `figures/src/fig_robust.py`.

Table 5: Sensitivity variants against the baseline analysis, part (a): margin of 1 and 2 points (`results/summary/summary_variants.csv`). $n$: targets; feasible: share of $n$ with first place reachable (in parentheses: baseline share for the same caps, targets and rule); cens.: exact solves that hit the time limit, counted neither as feasible nor infeasible; $L_1$ med.: median total points among feasible targets, in the variant's units; $n_{\mathrm{both}}$: targets feasible with positive cost under both; ratio: median per-target cost ratio variant/baseline. Generated by `figures/src/tables_ext.py`.

| variant, cap | rule | n | feasible (baseline) | cens. | $L_1$ med. | $n_{\mathrm{both}}$ | ratio |
|---|---|---|---|---|---|---|---|
| HELM, margin 1 pt, cap emp99T | mean | 78 | 0.15 (0.19) | 0 | 229.1 | 12 | 1.33 |
| HELM, margin 1 pt, cap emp99T | median | 78 | 0.28 (0.33) | 0 | 30.1 | 22 | 1.55 |
| HELM, margin 1 pt, cap emp99T | min | 78 | 0.12 (0.12) | 0 | 4.7 | 8 | 1.50 |
| HELM, margin 1 pt, cap emp99T | Borda | 78 | 0.19 (0.21) | 0 | 116.3 | 15 | 1.16 |
| HELM, margin 1 pt, cap emp99T | Condorcet | 78 | 0.32 (0.32) | 0 | 88.0 | 25 | 1.14 |
| HELM, margin 1 pt, cap emp99T | plurality | 78 | 0.53 (0.58) | 0 | 35.7 | 40 | 1.38 |
| HELM, margin 2 pt, cap emp99T | mean | 78 | 0.12 (0.19) | 0 | 270.3 | 9 | 1.73 |
| HELM, margin 2 pt, cap emp99T | median | 78 | 0.21 (0.33) | 0 | 38.8 | 16 | 2.34 |
| HELM, margin 2 pt, cap emp99T | min | 78 | 0.09 (0.12) | 0 | 6.7 | 6 | 2.26 |
| HELM, margin 2 pt, cap emp99T | Borda | 78 | 0.19 (0.21) | 0 | 117.2 | 15 | 1.17 |
| HELM, margin 2 pt, cap emp99T | Condorcet | 78 | 0.32 (0.32) | 0 | 90.2 | 25 | 1.17 |
| HELM, margin 2 pt, cap emp99T | plurality | 78 | 0.46 (0.58) | 0 | 43.4 | 35 | 1.95 |
| OLLB, margin 1 pt, cap emp99 | mean | 4495 | 0.11 (0.12) | 0 | 77.1 | 500 | 1.08 |
| OLLB, margin 1 pt, cap emp99 | median | 4495 | 0.09 (0.10) | 0 | 29.5 | 403 | 1.15 |
| OLLB, margin 1 pt, cap emp99 | min | 4495 | 0.04 (0.06) | 0 | 9.6 | 191 | 1.25 |
| OLLB, margin 2 pt, cap emp99 | mean | 4495 | 0.10 (0.12) | 0 | 82.0 | 466 | 1.17 |
| OLLB, margin 2 pt, cap emp99 | median | 4495 | 0.08 (0.10) | 0 | 32.2 | 341 | 1.34 |
| OLLB, margin 2 pt, cap emp99 | min | 4495 | 0.03 (0.06) | 0 | 10.9 | 115 | 1.53 |

Table 6: Sensitivity variants, part (b): OLLB with merged models removed, raw (un-normalised) task scores, and the top-500 slice of the board; columns as in Table 5. Raw-score rows with the $+10$ cap use raw points, so their ratio compares costs in different units. For the OLLB rank rules on the top-500 slice the baseline covers the 50 best targets of the full board only, so the baseline share is not comparable. The censored count (1) is one Borda target.

| variant, cap | rule | n | feasible (baseline) | cens. | $L_1$ med. | $n_{\mathrm{both}}$ | ratio |
|---|---|---|---|---|---|---|---|
| OLLB no merged models, cap emp99 | mean | 3788 | 0.11 (0.11) | 0 | 74.1 | 435 | 1.00 |
| OLLB no merged models, cap emp99 | median | 3788 | 0.09 (0.09) | 0 | 26.6 | 356 | 1.00 |
| OLLB no merged models, cap emp99 | min | 3788 | 0.07 (0.06) | 0 | 7.4 | 212 | 0.81 |
| OLLB no merged models, cap 100 | mean | 3788 | 1.00 (1.00) | 0 | 183.3 | 3788 | 1.00 |
| OLLB no merged models, cap 100 | median | 3788 | 1.00 (1.00) | 0 | 108.0 | 3787 | 1.00 |
| OLLB no merged models, cap 100 | min | 3788 | 1.00 (1.00) | 0 | 42.4 | 3787 | 0.93 |
| OLLB raw scores, cap +10 | mean | 4495 | 0.07 (0.03) | 0 | 48.5 | 113 | 0.74 |
| OLLB raw scores, cap +10 | median | 4495 | <0.01 (0.05) | 0 | 15.7 | 20 | 5.65 |
| OLLB raw scores, cap +10 | min | 4495 | 0.05 (0.10) | 0 | 12.6 | 217 | 1.50 |
| OLLB raw scores, cap 100 | mean | 4495 | 1.00 (1.00) | 0 | 154.1 | 4495 | 0.85 |
| OLLB raw scores, cap 100 | median | 4495 | 1.00 (1.00) | 0 | 92.9 | 4494 | 0.86 |
| OLLB raw scores, cap 100 | min | 4495 | 1.00 (1.00) | 0 | 74.7 | 4493 | 1.61 |
| OLLB top-500 slice, cap emp99 | mean | 499 | 0.95 (0.95) | 0 | 70.3 | 473 | 1.00 |
| OLLB top-500 slice, cap emp99 | median | 499 | 0.79 (0.79) | 0 | 25.5 | 395 | 1.00 |
| OLLB top-500 slice, cap emp99 | min | 499 | 0.40 (0.40) | 0 | 7.6 | 200 | 1.00 |
| OLLB top-500 slice, cap emp99 | Borda | 499 | 0.91 (0.10) | 1 | 36.5 | 48 | 0.91 |
| OLLB top-500 slice, cap emp99 | Condorcet | 499 | 0.30 (0.08) | 0 | 42.2 | 42 | 1.00 |
| OLLB top-500 slice, cap emp99 | plurality | 499 | 0.56 (0.07) | 0 | 29.3 | 36 | 0.70 |
| OLLB top-500 slice, cap 100 | mean | 499 | 1.00 (1.00) | 0 | 71.1 | 499 | 1.00 |
| OLLB top-500 slice, cap 100 | median | 499 | 1.00 (1.00) | 0 | 26.5 | 498 | 1.00 |
| OLLB top-500 slice, cap 100 | min | 499 | 1.00 (1.00) | 0 | 12.3 | 498 | 1.00 |
| OLLB top-500 slice, cap 100 | Borda | 499 | 1.00 (0.10) | 1 | 36.3 | 48 | 0.91 |
| OLLB top-500 slice, cap 100 | Condorcet | 499 | 1.00 (0.10) | 0 | 47.0 | 49 | 1.00 |
| OLLB top-500 slice, cap 100 | plurality | 499 | 1.00 (0.10) | 0 | 26.3 | 49 | 0.78 |

Table 7: Sensitivity variants, part (c): random task weights and cost in standard-deviation units; columns as in Table 5. For weights rows $n$ counts weight draws over the stratum targets, and the ratio is the weighted-mean cost over the equal-weight cost of the same target. SD-unit cost has no baseline: points are divided by the across-model standard deviation of each task (not by the evaluation-noise standard error).

| variant, cap | rule | n | feasible (baseline) | cens. | $L_1$ med. | $n_{\mathrm{both}}$ | ratio |
|---|---|---|---|---|---|---|---|
| HELM, weights, cap emp99T | Dirichlet(1) | 3000 | 0.68 (0.67) | 0 | 33.6 | 1739 | 0.43 |
| HELM, weights, cap emp99T | Dirichlet(5) | 3000 | 0.67 (0.67) | 0 | 74.9 | 1836 | 0.68 |
| OLLB, weights, cap emp99 | Dirichlet(1) | 3500 | 0.91 (1.00) | 0 | 12.2 | 3128 | 0.54 |
| OLLB, weights, cap emp99 | Dirichlet(5) | 3500 | 0.99 (1.00) | 0 | 19.1 | 3440 | 0.70 |
| OLLB, weights, cap emp99 | one-hot | 42 | 0.33 (1.00) | 0 | 8.5 | 14 | 0.38 |
| HELM, SD-unit cost, cap emp99T | mean | 78 | 0.19 (–) | 0 | 17.5 | – | – |
| HELM, SD-unit cost, cap emp99T | median | 78 | 0.33 (–) | 0 | 2.5 | – | – |
| HELM, SD-unit cost, cap emp99T | min | 78 | 0.12 (–) | 0 | 0.6 | – | – |
| HELM, SD-unit cost, cap 100 | mean | 78 | 1.00 (–) | 0 | 52.0 | – | – |
| HELM, SD-unit cost, cap 100 | median | 78 | 1.00 (–) | 0 | 10.2 | – | – |
| HELM, SD-unit cost, cap 100 | min | 78 | 1.00 (–) | 0 | 8.3 | – | – |
| OLLB, SD-unit cost, cap emp99 | mean | 4495 | 0.12 (–) | 0 | 4.3 | – | – |
| OLLB, SD-unit cost, cap emp99 | min | 4495 | 0.06 (–) | 0 | 1.6 | – | – |
| OLLB, SD-unit cost, cap 100 | mean | 4495 | 1.00 (–) | 0 | 11.1 | – | – |
| OLLB, SD-unit cost, cap 100 | min | 4495 | 1.00 (–) | 0 | 6.4 | – | – |

Table 8: Paper figure or table, generating command and input.

| output | command (`python3 figures/src/...`) | input under `R` |
|---|---|---|
| Fig. 1 | `fig_rank_rules.py R ollb/helm figures/F_{ollb,helm}_primary -caps primary` | `core_{ollb,helm}.csv` |
| Fig. 2 | `fig_scaling.py R figures/F_helm_scaling` | `summary/summary_rob_helm_tasksub_emp99T.csv` |
| Table 1, 2 | `tables.py R tables` (T4, T3) | `core_*.csv`, `summary/frontier_*.csv` |
| Table 3 | `tables.py R tables` (T2) | `theory_check.csv` |
| Table 4 | `tables_ext.py R tables` (T5) | `summary/frontier_helm.csv` |
| Fig. 3 | `fig_rank_rules.py ...  -caps ceil` | `core_*.csv` |
| Fig. 4, 5 | `fig_cap_grid.py R ollb/helm figures/F_{ollb,helm}_capgrid` | `core_*.csv` |
| Fig. 6 | `fig_robust.py R figures/F_robust` | `summary/summary_rob_*.csv`, `core_*.csv` |
| Tables 5–7 | `tables_ext.py R tables` (T6) | `summary/summary_variants.csv` |

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